

CONFIDENTIAL

2022

D12000i RATO Technology Product Introduction

Precautions



- **Purpose:**

- Ensure the health, life safety, property safety and environmental protection of operators during daily maintenance and repair, and secure that the operation is safe and effective;

- **Safety regulations:**

- 1. The generator shall be used in a **cool and ventilated** place;
- 2. It is strictly prohibited to **smoke** around the generator during operation, and make sure **there are no combustibles around**;
- 3. In case of rainstorm and snowy weather, stop the generator immediately;
- 4. When the generator is in operation, make sure there is no object within 2 meters around which might hinder the heat dissipation and ventilation of the generator;
- 5. It is prohibited to add gasoline to the oil tank during the operation of the generator;
- 6. It is forbidden to **charge with the generator on the vehicle**;
- 7. If it is not used for more than 30 days, it is necessary to drain gasoline and engine oil in advance;

- **I. Environmental safety:**

- 1. To ensure personal and property safety, **fire extinguishers** must be equipped within 2 meters of the operation scope;
- 2. **It is prohibited to charge with generator indoors** as there might be risk of carbon monoxide poisoning;
- 3. The maintenance personnel must confirm the surrounding environment before operation to ensure that there is **no open fire source, the air is well ventilated, and the light is sufficient**;
- 4. After disassembly, check whether there is fuel and engine oil **leakage**;
- 5. It is strictly prohibited to expose fuel to the air, and it must be stored in sealed containers;
- 6. It is strictly forbidden to dump the engine oil anywhere. It should be collected and delivered to special treatment organization for disposal;

- **II. Operation safety:**

- 1. **It is strictly prohibited to work after drinking, smoke** during operation, **and work in poor mental state**;
- 2. It is strictly prohibited to wear sandals, slippers and other shoes without toe protection during operation;
- 3. Strictly use labor protection articles (gloves, masks, earplugs, etc.);
- 4. It is prohibited to use unqualified tools to disassemble and assemble the machine;
- 5. It is prohibited to verify the durability of the generator without personal care;
- 6. It is prohibited to adjust the parameters of the generator at will;
- 7. It is prohibited to use parts not authorized or sold directly by our company for repair and maintenance;

- **Generator component introduction**
- **Generator exploded Views, illustrated books and circuits**
- **Generator circuit test methods**
- **Generator maintenance manual**
- **Generator fault lights**
- **Generator requirements**

Generator introduction

Oil pump position

ECU

**Primary
filter**

Fuel drain hose

Charger

Secondary air filter



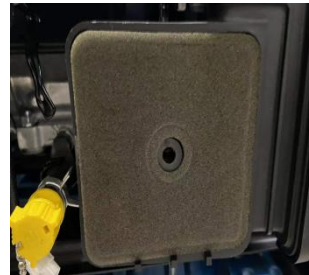
Generator introduction



Generator introduction



Oxygen sensor



Primary filter element



Secondary filter element



ECU



Oil passage nozzle



Fuel pump



Inverter/DC controller
Left: DC controller
Right: inverter



Ignition coil

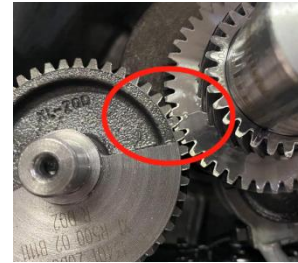
Generator introduction



Piston/connecting rod:
confirmation point
Left: **connecting rod**
Right: **piston**



Cam point



**Horizontal axis
point**



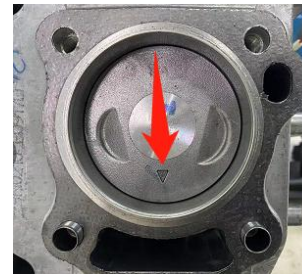
Valve assembly
Left: inlet valve (IN)
Right: exhaust valve
(EX)



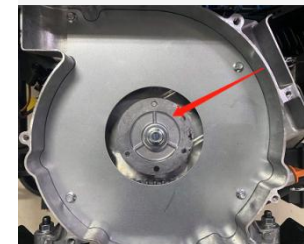
Trigger



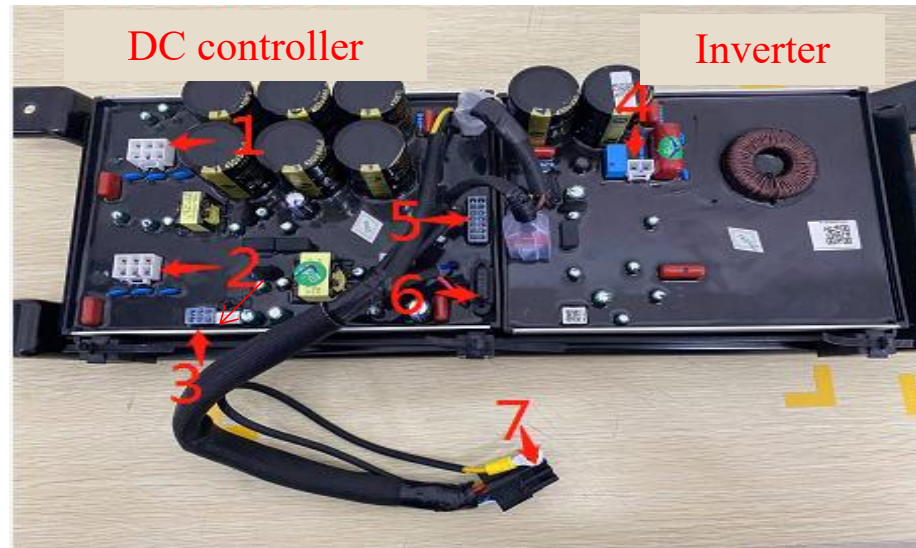
Rocker arm



**Piston assembly
point**

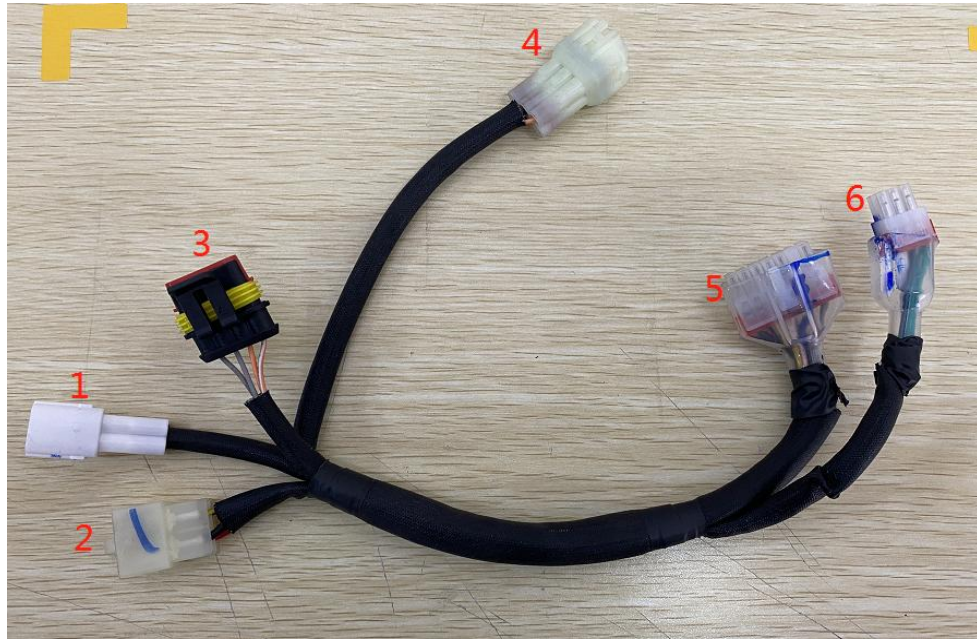


**Impeller mounting
plate**



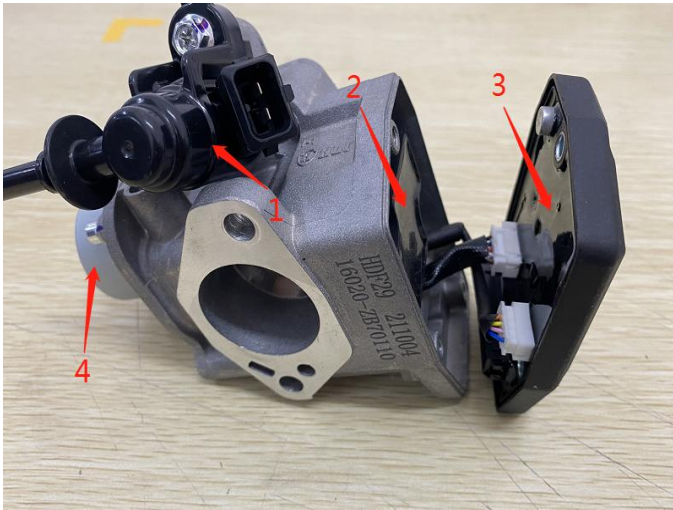
Component introduction

- ① Connect with magneto harness- detect any two groups of coils **AC (330V-420V)**
- ② Connect with magneto harness- detect any two groups of coils **AC (330V-420V)**
- ③ Connect with the engine harness connector; ECU power supply line-**DC-12V, detect the lower right two sockets**
- ④ Panel **1.5 KW** output line connection - **AC (230V)**
- ⑤ Electric starting lines - two red and two black, and DC controller input line
- ⑥ Electric starting fuse
- ⑦ Charger plug - detect black/yellow line **DC (720V-740V)**



Component introduction

- ① Connected to ECU circuit - **DC-12V** voltage is supplied to ECU module
- ② Connected to starting motor circuit - DC voltage is supplied from battery- **(50V-60V)**
- ③ Magneto output circuit (2 gray) **35V-AC**, output to power supply circuit of DC controller
- ④ Connected with panel harness - control start and flameout harness
- ⑤ Connected with DC controller, electric starting line, magneto power supply line and flameout harness
- ⑥ DC control output to ECU circuit - output **DC-12V**



Component introduction

- ① Nozzle-gasoline circuit
- ② Electronic control module - combined with ECU
- ③ ECU Circuit Program Module
- ④ Throttle stepper motor

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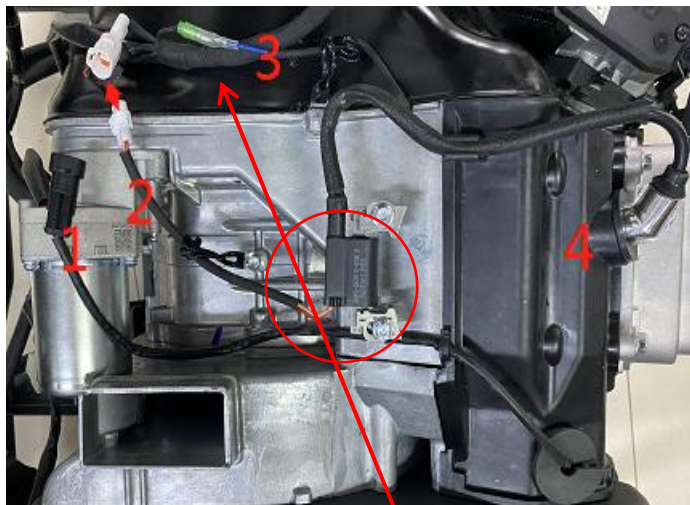
Fuel pump



Component introduction

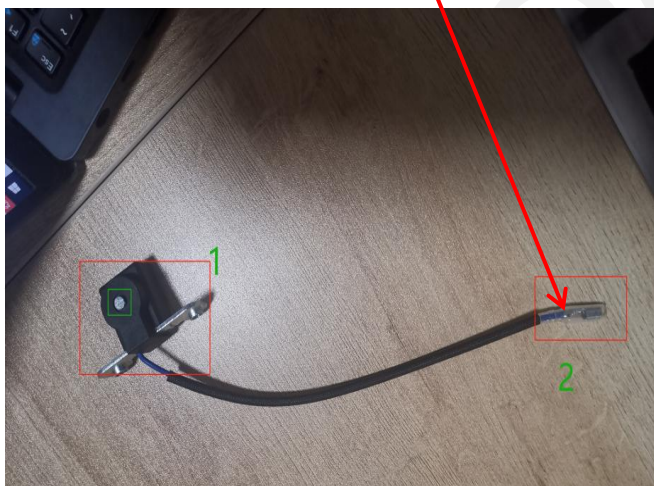
- ① ECU power supply line - **12V-DC**
- ② Oil outlet nozzle - gasoline nozzle port
- ③ Oil pump motor
- ④ Fuel filter

Ignition coil and trigger



Component introduction

- ① Wire speed of oxygen sensor;
- ② ECU power supply line, pay attention to prevent leakage at **high voltage**
- ③ Trigger detect resistance - **(300 ± 50)**
- ④ Spark plug cap - **high voltage** output, pay attention when checking



Component introduction

- ③ Trigger; detect resistance - **(300 ± 50)**
- ② Connected with the harness (blue harness);

Test data meeting normal charging requirements of generator

- 1: Charger plug DC-DC voltage 720V-740V
- 2: Output voltage of magneto AC-AC 430V-460V
- 3: Output voltage of magneto AC-AC 430V-460V

If all the above voltage value requirement could be met, the generator will not charge the battery, and the DC power supply will not work, then the DC power supply is faulty.



Generator panel

Panel output request confirmation

1: DC controller output to inverter jack line (1-3)

2: Inverter jack line output voltage - 230V (4)



**DC controller output-1 DC-
360V-460V**



**DC controller output - 2
DC-360V-460V**



**DC controller output - 3 DC-
710V-740V**



**Inverter output-4
AC-230V**

Generator electric starting

- Electric starting requirements

1: Turn on the aircraft battery and connect it to the generator charger

2: The fuse on the DC controller is not damaged

3: The tested voltage in the DC control plug is (50V-60V) DC voltage - as shown in Figure 1

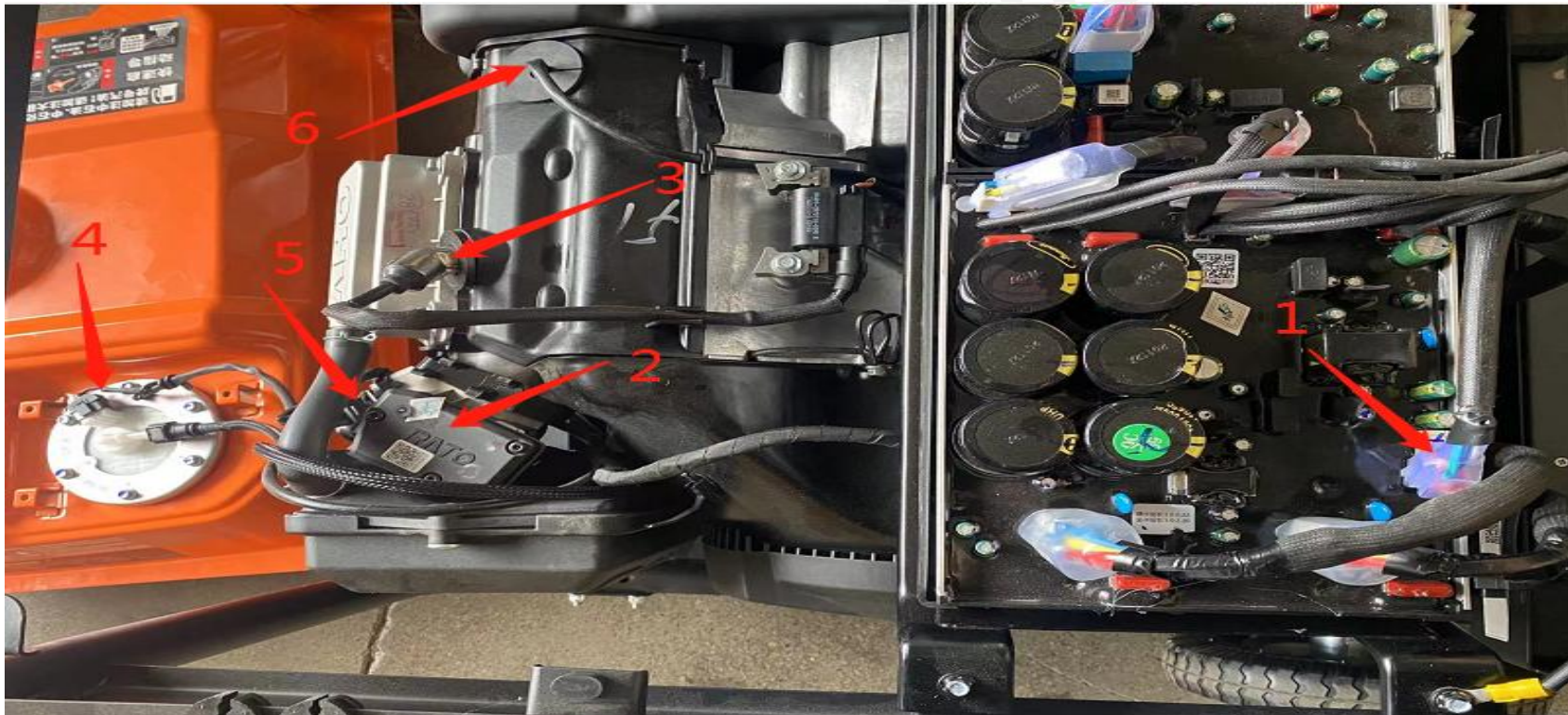
4: The voltage of relay connecting harness is synchronous with the battery voltage (50V-60V)



ECU Power Supply Mode



1. The power of ECU (2) is supplied by No. 1 circuit(12V DC), the red light should be on when ECU starts, and be off after ECU startup successfully
2. The ECU module controls the power supply to (oil pump - 4, nozzle - 5, ignition coil - 3) respectively
3. ECU controls supplies power to ignition coil through circuit No. 3
4. ECU controls power supply to oil pump ---- **12V** DC voltage through circuit No. 4
5. ECU controls power supply to nozzle ---- **12V** DC voltage through circuit No.5
6. Circuit No. 6 supplies power to oxygen sensor ---- **the altitude response regulation is fed back to** ECU module and controls the engine operation



Maintenance Manual



★ Basic maintenance of parts

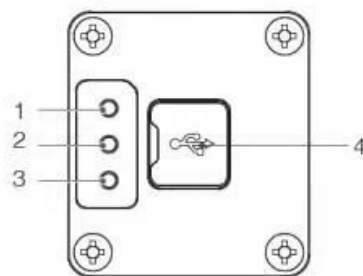
The charging station shall be inspected and maintained regularly to ensure it is in good condition. The charging station must be closed before maintenance. For maintenance items not covered in this document, please contact DJI or an authorized dealer of DJI.

Item		Each time	The 1st month or 20 hours	Every 3 months or 50 hours	Every 6 months or 100 hours	Every year or every 300 hours
Engine oil	Check the oil level*	√				
	Replacement		√	√		
Air filter element	inspection	√				
	Cleaning and replacement			√		
Spark plug	inspection				√	
	Cleaning and replacement					√
Spark arrester	inspection			√		
	Cleaning and replacement				√	
Fuel pipe	inspection	Every two years (replace if necessary)				

Generator fault lights

		Description
1. Channel light	ON	Wait for the battery to plug in
	Flash	Battery is charging
	Shell	Battery is fully charged
	Flash	Battery or channel alarm
	Flash	Battery or channel failure
2. Charger	Flash	Charger alarm
	Flash	Charger fault
3. Generator	Flash	Charger alarm
	Flash	Generator fault
	Flash 4	Maintenance reminding

Status lamp



1. Charging channel status lamp
2. DC module status lamp
3. Generator set/AC module status lamp
4. Small USB-C interface protective cover

Status lamp	Flashing mode	Description
Charging channel status lamp	Yellow light is always on	Wait for the battery to plug in
	Green light is flashing	Battery is charging
	Green light is always on	Battery is fully charged
	Yellow light is flashing	Battery or channel alarm
	Red light is flashing	Battery or channel failure
DC module status lamp	Yellow light flashes 4 times	DC module not active
	Yellow light is flashing	DC module alarm
	Red light is flashing	DC module fault
Generator set/AC module status lamp	Yellow light flashes 4 times	Generator set/AC module not activate
	Yellow light is flashing	Generator set/AC module alarm
	Red light is flashing	Generator set/AC module fault
	Red light flashes 4 times	Generator set maintenance reminding

Generator requirements



I. Generator activation process

1. Check the generator appearance for abnormal damages
2. Add engine oil and gasoline, start the generator, connect the remote control of the aircraft and wait for 10 seconds
3. **Do not activate the generator without starting it**
4. If it has already been activated and is not activated according to the activation sequence, the generator needs to be restarted. Connect the remote controller to the activation interface and wait for 10 seconds. Ask the after-sales service personnel of Dajiang for the specific operation process

I: Precautions for handling customer feedback

1. Confirm the types of customer feedback questions - they can be divided into the following 5 categories

Unable to start: Check whether the ECU module is powered on. The ECU light must be red once with a beep when starting. If there is no beep, the ECU is faulty
Check the spark plug ignition, engine oil, gasoline, valve clearance, cylinder pressure, trigger resistance, and DC exist in the wire connected to the igniter-above 8V is qualified.
Disconnect the oil sensor and flame out for fault confirmation firstly (for **T20P**)

Unable to charge of generator: When the motor is working, check whether there is a voltage between the magneto output and the charger plug.
If the charger plug has a voltage, the charger is faulty. (Contact the after-sales service personnel of Dajiang for troubleshooting)

Unstable generator operation: Check whether the DC power supply and ECU have a fault light. If there is a fault light, solve the problem according to the fault light, or upgrade the generator to version 421. Check the gasoline circuit, valve clearance, fixing nuts and gaskets of the air filter

No output from panel AC socket: Check magneto output and test inverter voltage

Poor starting of generator: Check whether the battery charging port is open, check the fuse and relay power supply circuit